

## STUDY AND LEARNING CENTRE

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### STUDY TIPS

## PME1.7: RULES FOR SIGNIFICANT FIGURES

In scientific measurements, significant figures are used to indicate the accuracy of a measurement. The last digit in a measurement is often an estimate, a good guess.

All the digits in a measured value, including the last estimated digit, are called significant figures or significant digits.

Consider a series of measurements made with (a) an old wooden ruler without mm marks, (b) a more accurate steel ruler, (c) steel callipers with a Vernier scale.

|     | Reported Result | Number of Significant Figures in the measurement | Accuracy of measurement                           | Implied range of possible values |
|-----|-----------------|--------------------------------------------------|---------------------------------------------------|----------------------------------|
| (a) | 7 cm            | 1                                                | To the nearest 1 cm,<br>ie $7 \pm 0.5$ cm         | 6.5 cm – 7.5 cm                  |
| (b) | 7.2 cm          | 2                                                | To the nearest 0.1 cm,<br>ie $7.2 \pm 0.05$ cm    | 7.15 – 7.25 cm                   |
| (c) | 7.23            | 3                                                | To the nearest 0.01 cm,<br>ie $7.23 \pm 0.005$ cm | 7.225 – 7.235 cm                 |

### The number of Significant Figures in a measured value

**Rule 1** Any non-zero digit is significant. The position of a decimal point makes no difference.

|                  |        |   |           |
|------------------|--------|---|-----------|
| <i>Example 1</i> | 15.7   | 3 | sig. figs |
|                  | 157    | 3 | sig. figs |
|                  | 1.57   | 3 | sig. figs |
|                  | 2.7942 | 5 | sig. figs |

**Rule 2** Zeros between numbers **are** significant.

|                  |         |   |           |
|------------------|---------|---|-----------|
| <i>Example 2</i> | 1.05    | 3 | sig. figs |
|                  | 10.51   | 4 | sig. figs |
|                  | 200.708 | 6 | sig. figs |

**Rule 3** Zeros at the right hand end of whole numbers are **not** significant, unless otherwise stated.

|                  |       |   |           |
|------------------|-------|---|-----------|
| <i>Example 3</i> | 70    | 1 | sig. figs |
|                  | 2860  | 3 | sig. figs |
|                  | 15090 | 4 | sig. figs |

**Rule 4** Zeros at the left hand end of decimal numbers are **not** significant.

|                  |        |   |           |
|------------------|--------|---|-----------|
| <i>Example 4</i> | 0.28   | 2 | sig. figs |
|                  | 0.0039 | 2 | sig. figs |
|                  | 0.0604 | 3 | sig. figs |

**Rule 5** Zeros at the right hand end of decimal numbers **are** significant.

|                  |         |   |           |
|------------------|---------|---|-----------|
| <i>Example 5</i> | 12.0    | 3 | sig. figs |
|                  | 0.760   | 3 | sig. figs |
|                  | 0.48300 | 5 | sig. figs |
|                  | 2.07090 | 6 | sig. figs |

### Significant Figures and Scientific Notation

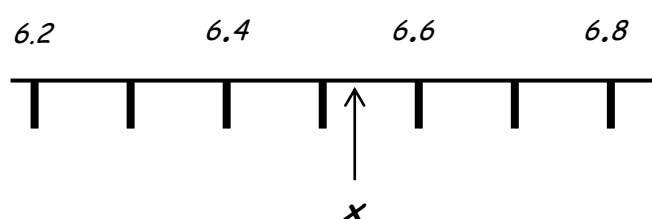
The problems of deciding how many significant figures a value has is simplified by writing its value in Scientific Notation.

Examine the number of digits in the first number below, not in the power of ten.

- |     |                                   |            |        |
|-----|-----------------------------------|------------|--------|
| (a) | $0.0003 = 3 \times 10^{-4}$       | 1 sig fig  | Rule 4 |
| (b) | $720000 = 7.2 \times 10^5$        | 2 sig figs | Rule 3 |
| (c) | $660 = 6.6 \times 10^2$           | 2 sig figs | Rule 3 |
| (d) | $660.0 = 6.600 \times 10^2$       | 4 sig figs | Rule 5 |
| (e) | $0.66000 = 6.6000 \times 10^{-1}$ | 5 sig figs | Rule 5 |
| (f) | $808.01 = 8.0801 \times 10^2$     | 5 sig figs | Rule 2 |

### Recording measurements and Significant Figures

Consider the scale below.



The measurement  $x$  is:

- Certainly greater than 6 and less than 7
- Certainly greater than 6.5 and less than 6.6
- Very probably greater than 6.53 and less than 6.55

The result would be recorded as 6.54 - a value with 3 significant digits. The first two digits are certain and the last is a good estimate.

Writing "6.54" implies " $6.54 \pm 0.005$ ", ie between 6.535 and 6.545, unless otherwise stated.

### Exercise

Write each of the following measurements in Scientific Notation and state the number of significant digits in the value.

- |             |            |             |              |
|-------------|------------|-------------|--------------|
| (a) 345     | (b) 17642  | (c) 0.0033  | (d) 0.000306 |
| (e) 870     | (f) 20000  | (g) 140.600 | (h) 710.0    |
| (i) 0.04080 | (j) 0.0050 |             |              |

### Answers

#### Exercise

- |              |                        |            |
|--------------|------------------------|------------|
| (a) 345      | $3.45 \times 10^2$     | 3 sig figs |
| (b) 17642    | $1.7642 \times 10^4$   | 5 sig figs |
| (c) 0.0033   | $3.3 \times 10^{-3}$   | 2 sig figs |
| (d) 0.000306 | $3.06 \times 10^{-4}$  | 3 sig figs |
| (e) 870      | $8.7 \times 10^2$      | 2 sig figs |
| (f) 20000    | $2 \times 10^4$        | 1 sig fig  |
| (g) 140.600  | $1.40600 \times 10^2$  | 6 sig figs |
| (h) 710.0    | $7.100 \times 10^2$    | 4 sig figs |
| (i) 0.04080  | $4.080 \times 10^{-2}$ | 4 sig figs |
| (j) 0.0050   | $5.0 \times 10^{-3}$   | 2 sig figs |